A COURSE-BASED PROJECT

On

FLIGHT DATA ANALYSIS

Submitted in partial fulfillment of Big Data Analytics

GRIET Lab On Board(G-LOB)

By

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CERTIFICATE

This is to certify that the GLOB entitled "**FLIGHT DATA ANALYSIS**" is submitted by **G. Akshara (23241A6724), K. Vasavi Chandana (23241A6727) and K. Shreya Shree (23241A6730)** in partial fulfillment of the award of degree in BACHELOR OF TECHNOLOGY in Computer Science and Engineering (Data Science) during Academic year 2025-2026.

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ABSTRACT

This project presents an analytical study of flight delay data using Apache Spark to investigate whether airline carriers can recover lost time during flight. Utilizing Spark's scalable distributed computing capabilities, the analysis processes more than 123 million flight records spanning 22 years. A central metric, "gain"—calculated as the difference between departure and arrival delays—is introduced to measure the time recovered or lost during flights. The dataset is filtered and enriched with airline metadata to build a robust model. A linear regression model is trained using key features such as flight distance, departure delay, and airline carrier to predict time gain. The analysis reveals that while some carriers show a slight edge in recovering time, the overall differences are small, with flight distance emerging as the most significant predictor. The results are validated and visualized through decile analysis and comparative plots. This project demonstrates the effectiveness of Apache Spark and machine learning in extracting meaningful operational insights from large-scale transportation data, with implications for airline efficiency and scheduling optimization.

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CHAPTER 1
INTRODUCTION

1.1 INTRODUCTION TO PROJECT WORK

Flight delays represent a major concern for passengers, airlines, and airport authorities, often leading to missed connections, increased operational costs, and customer dissatisfaction. One intriguing aspect of air travel is the phenomenon where some flights manage to "make up" for lost time while airborne, arriving closer to their scheduled time despite late departures. This project investigates that phenomenon using large-scale flight data and the distributed computing capabilities of Apache Spark. With over 123 million flight records spanning 22 years, the dataset provides a rich foundation for analyzing trends and behaviors across airline carriers.

The core objective of this analysis is to examine whether certain carriers consistently gain more time in flight than others, and which factors most significantly influence that recovery. To achieve this, the project utilizes Spark SQL for efficient querying and transformation of massive datasets, as well as Spark MLlib to build predictive models based on key flight attributes such as distance, departure delay, and carrier identity. By modeling and visualizing these trends, the study identifies which variables contribute most to time gain, evaluates the reliability of different airlines, and highlights areas where improvements in scheduling or logistics might be possible. This work not only demonstrates the power of big data analytics in aviation but also provides actionable insights that could lead to better-informed operational strategies and enhanced passenger experiences.

1.2 SIGNIFICANCE OF THE PROJECT

This project holds substantial significance in the realm of big data analytics, transportation planning, and operational efficiency in the airline industry. With the ever-increasing volume of air travel globally, understanding and optimizing flight operations has become both a challenge and a priority for airlines, airport authorities, and passengers alike. Flight delays are not only a major source of customer dissatisfaction but also contribute to logistical disruptions, fuel inefficiencies, and increased operational costs. This project addresses a key aspect of this issue: the ability of airlines to recover lost time during flights, a factor that is not often analyzed in standard delay reports.

By utilizing Apache Spark and Spark SQL, the project demonstrates how large-scale, distributed computing tools can be effectively applied to massive datasets—in this case, over 123 million flight records spanning two decades. Spark's parallel processing capabilities enable efficient handling of complex queries and machine learning tasks, which would otherwise be computationally intensive. The project's use of predictive modeling to assess the extent to which carriers can make up time in flight provides a data-driven perspective on airline performance, going beyond traditional metrics like on-time arrivals or average delays.

Moreover, the insights drawn from this analysis have practical value for multiple stakeholders. Airlines can leverage these findings to refine their scheduling and contingency planning strategies. Airport operations can better prepare for variability in arrival times, especially from long-distance flights. Additionally, the project lays the groundwork for further studies in predictive maintenance, route optimization, and passenger experience improvements. It also underscores the growing role of data science and big data platforms in addressing real-world industry problems, highlighting the importance of scalable tools like Spark in transforming raw transportation data into actionable insights.

CHAPTER 2
LITERATURE SURVEY

1. Analysis of Airline On-Time Performance Using Big Data Frameworks

Existing Approach: Chandrasekaran et al. (2016) conducted an analysis of airline performance using Hadoop-based frameworks, focusing on delay prediction and performance metrics. Their system processed millions of flight records to identify trends and anomalies in airline punctuality.
Drawback: While effective for batch processing, Hadoop lacked the iterative computing capabilities and in-memory processing offered by Spark, leading to slower performance for tasks such as machine learning and real-time analysis.

2. Predicting Flight Delays Using Machine Learning Models

Existing Approach: Ball et al. (2010) explored predictive modeling for flight delays using linear regression, decision trees, and support vector machines. The models used historical weather, traffic, and schedule data to forecast potential delays.
Drawback: The models were implemented on traditional systems and did not scale well with increasing data volume. Additionally, they often failed to generalize across different airports or time periods due to overfitting or lack of feature engineering.

3. Time Series Analysis of Flight Delays

Existing Approach: Ahmed and Raffa (2014) applied ARIMA models to study seasonal patterns in flight delays using time series analysis. Their findings provided valuable insights into temporal delay trends at major US airports.
Drawback: ARIMA models are limited to univariate analysis and assume stationarity. They also struggle with handling large-scale datasets and multidimensional input features, reducing their applicability in real-world airline delay analytics.

4. Airline Data Mining with Spark SQL

Existing Approach: Natarajan et al. (2018) utilized Apache Spark SQL to query and analyze large airline datasets stored in distributed file systems. Their study highlighted the benefits of Spark's distributed querying for aggregating delay statistics and filtering relevant records.
Drawback: The approach primarily focused on descriptive analytics and lacked advanced machine learning integration. Insights were limited to summary statistics without predictive

modeling or actionable outcome generation.

5. Evaluating Carrier Performance Using Flight Data Visualization

Existing Approach: Xu and Wang (2017) implemented a dashboard-based visualization system for evaluating airline performance using Tableau and preprocessed CSV files. The system enabled users to explore flight delays, carrier statistics, and route efficiencies interactively.

Drawback: The system relied heavily on pre-aggregated data and lacked real-time capabilities. Without a big data backend, it struggled to scale for interactive exploration of multi-million-row datasets, limiting its utility in live environments.

CHAPTER 3
PROPOSED METHOD

3.1 PROBLEM STATEMENT

Air travel has become an essential mode of transportation in the modern world, but it is frequently hampered by delays that can cause significant disruptions for passengers, airlines, and airport operations. Delays not only lead to economic losses and logistical complications but also negatively impact customer satisfaction and airline reputations. While some delays are caused by uncontrollable external factors such as weather or air traffic congestion, others result from inefficiencies in airline scheduling and operations. Interestingly, some airline carriers are known to "make up time" during flights—recovering part of the delay between departure and arrival.

With over two decades of publicly available flight data amounting to more than 120 million records, there exists a valuable opportunity to explore delay patterns at scale using advanced big data tools. Traditional data analysis methods fall short when it comes to handling this volume and velocity of data, making it essential to adopt scalable, distributed computing frameworks like Apache Spark and Spark SQL.

This project aims to address the problem of understanding and predicting flight delay recovery by analyzing historical flight data through Spark-based processing and machine learning techniques. Specifically, it investigates whether certain airline carriers are more effective at recovering time lost during delayed departures, and how variables like flight distance and initial delay affect the potential to gain time in the air. The core goal is to develop a predictive model that estimates the time gained (or lost) during flight, using structured flight data and distributed computation.

By providing actionable insights into carrier performance and the factors influencing flight delay recovery, this project seeks to support data-driven decision-making for airline operations, customer planning, and policy formulation. Ultimately, it contributes to enhancing transparency and efficiency in the aviation industry by harnessing the power of big data analytics.

3.2 OBJECTIVES OF THE PROJECT

To preprocess and analyze large-scale flight data

Utilize Apache Spark and Spark SQL to efficiently load, clean, and process over 120 million flight records spanning multiple years, handling challenges related to data volume, missing values, and inconsistencies.

To investigate patterns in flight delays and recoveries

Identify and quantify the extent to which airlines are able to recover lost time during flights, and examine how this varies across carriers, routes, and flight durations.

To build a predictive model for time gained in-flight

Develop and evaluate a linear regression model to predict the difference between departure delay and arrival delay (i.e., time gained) based on features such as flight distance, airline carrier, and departure delay.

To compare airline performance in delay recovery

Rank airline carriers by their average time recovery and analyze whether certain carriers consistently outperform others in compensating for delayed departures.

To visualize insights for actionable decision-making

Generate visualizations such as gain vs. prediction plots and decile-based performance charts to help stakeholders (e.g., airlines, passengers, analysts) understand model outputs and performance patterns.

To demonstrate the effectiveness of Spark for big data analytics

Highlight the scalability and efficiency of Apache Spark in handling real-world datasets and machine learning workflows in a distributed computing environment.

3.3 ARCHITECTURE DIAGRAM

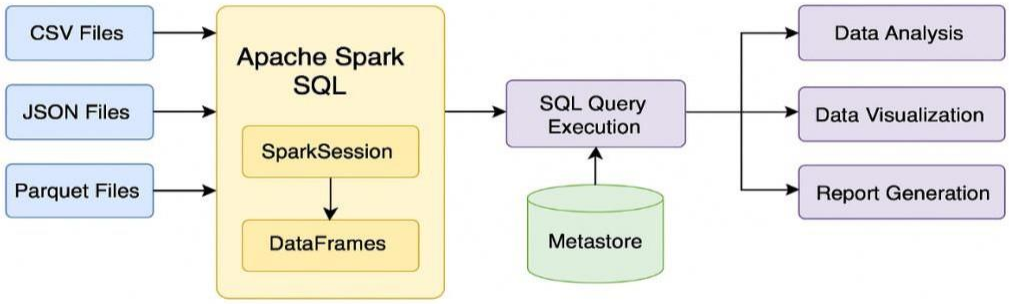

Fig:1

1. Data Sources

The architecture begins with the ingestion of structured data from various file formats commonly used in big data environments. These include CSV (Comma-Separated Values) files, JSON (JavaScript Object Notation) files, and Parquet files. Each of these formats offers different advantages in terms of readability, storage efficiency, and compatibility with analytical tools. Apache Spark is capable of handling all these formats seamlessly, enabling users to process data from diverse sources without requiring manual transformation. This flexibility ensures that data from real-time logs, transactional systems, or third-party APIs can be efficiently brought into the processing pipeline.

2. Apache Spark SQL

At the heart of the architecture lies Apache Spark SQL, which is responsible for managing and processing the incoming data. Using the SparkSession, which acts as the main entry point for all Spark functionalities, the system loads the data into a structure known as a DataFrame. DataFrames are distributed collections of data organized into named columns, much like a relational database table. This tabular abstraction allows for efficient execution of SQL queries and integration with Spark's machine learning and graph processing libraries. Spark SQL bridges the gap between declarative SQL-style querying and complex data transformation tasks, allowing users to write sophisticated queries with minimal code.

3. SQL Query Execution and Metastore

Once the data is loaded into DataFrames, it undergoes SQL-based querying and transformation through Spark SQL's query execution engine. During this stage, queries are optimized and executed across a distributed computing cluster. The architecture includes a Metastore component, typically implemented using the Hive Metastore, which stores metadata about the data such as table names, column types, and partition information. This metadata plays a crucial role in optimizing query performance and managing schema consistency, especially when dealing with large-scale datasets that span multiple files and formats. The Metastore also enables schema evolution and supports SQL operations like joins and aggregations across multiple datasets.

4. Output and Utilization

After the SQL queries are executed and results are generated, the final stage of the architecture focuses on using these results for various analytical purposes. One key application is data analysis, where the processed results are examined to uncover patterns, correlations, and business insights. Additionally, the data is visualized using charts, graphs, or dashboards to support intuitive interpretation and decision-making. The processed data may also be compiled into structured reports for regular monitoring, performance tracking, or presentations to stakeholders. Together, these outputs make the architecture suitable for end-to-end analytics, enabling organizations to move from raw data ingestion to actionable intelligence.

Software Requirements

This Spark SQL-based flight data analysis project uses a local, standalone Spark environment. The required software includes:

- **Apache Spark (v3.x)**: The main distributed computing engine used to process and analyze large-scale flight data using SQL queries.
- **Hdfs**: Distributed storage system for managing large datasets.
- **Python (v3.7 or higher)**: The core programming language used to write Spark code through the PySpark API.
- **PySpark Library**: The Python API for Apache Spark, which allows Python scripts to interface with Spark’s capabilities including SQL queries and DataFrames.
- **Jupyter Notebook**: An interactive development environment used to write, test, and visualize the Spark SQL code and outputs.

- **Data Source Files**: CSV files (such as flights.csv and airlines.csv) that are loaded into Spark DataFrames for querying and analysis.
- **Visualization Libraries**: Python libraries like matplotlib and seaborn are optionally used for graphing and visualization of analytical results.

The project does not rely on a Hadoop Distributed File System (HDFS), YARN, or Hive Metastore. All processing is done locally using the built-in Spark SQL engine.

Hardware Requirements

Because the dataset size is relatively moderate and the project runs in a local Spark environment, the hardware requirements are modest:

- **Processor**: Dual-core or quad-core CPU (Intel i5 or equivalent minimum; i7 or Ryzen 7 recommended for faster performance)
- **Memory (RAM)**:
  - **Minimum**: 4 GB
  - **Recommended**: 8 GB or higher, especially if running multiple notebooks or handling larger CSV files
- **Storage**:
  - **Minimum**: 2–5 GB free disk space (for data files, Python environment, and Spark installation)
  - **Recommended**: SSD over HDD to improve data load and query speed
- **Operating System**: Any OS compatible with Spark and Python (Linux, macOS, Windows)

3.4 MODULES CONNECTIVITY DIAGRAM AND ITS DESCRIPTION

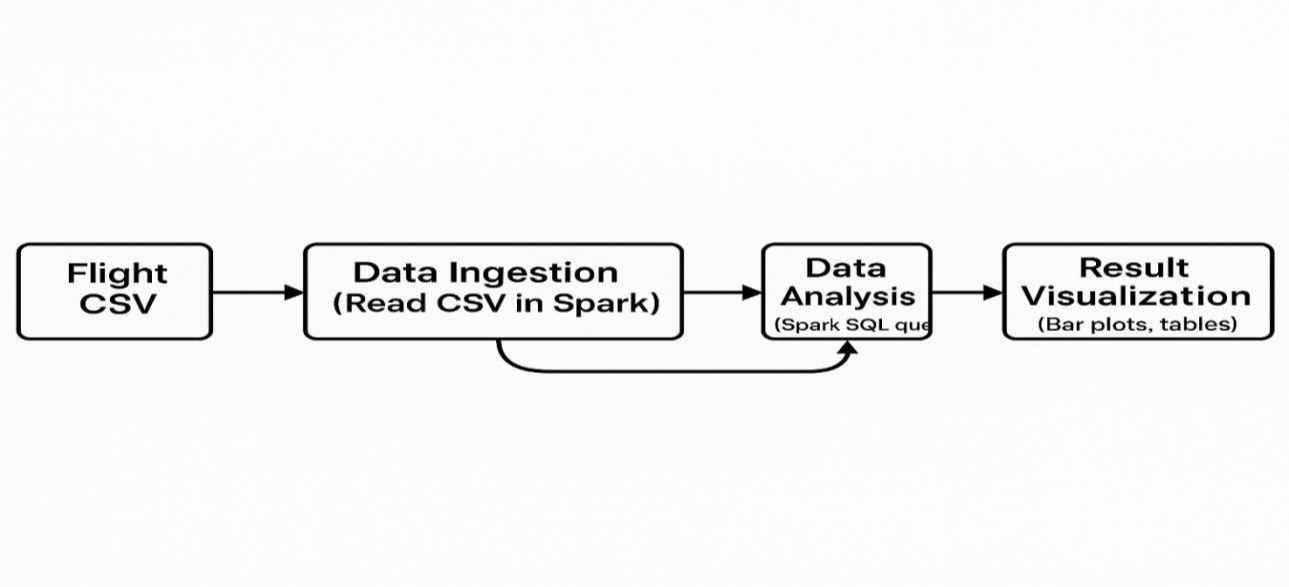

Fig:2

The module connectivity diagram for the Spark SQL flight data analysis project represents the logical interaction between various functional modules used throughout the project pipeline. This structured flow outlines how data travels and is transformed from its raw format into insightful analysis.

Flight CSV Module

This module represents the raw data input in CSV format. It serves as the foundational source of information containing historical flight records. These files typically include fields like flight number, origin and destination airports, departure and arrival times, delay durations, and carrier codes.

Data Ingestion Module (Read CSV in Spark)

Once the flight data CSV files are available, the Data Ingestion module uses Apache Spark to load this data. Spark's distributed processing capability enables it to handle large-scale datasets efficiently. The ingestion step uses SparkSession.read.csv() to parse and convert the raw data into structured DataFrames, making it suitable for further processing and querying.

Data Cleaning & Preprocessing Module

This module ensures the dataset is refined for analysis. It handles missing values, filters out invalid or irrelevant records (e.g., null delay values or outliers), and may perform column transformations. The preprocessing also includes selecting a subset of data based on conditions

such as years or delay thresholds, and joining with lookup tables like airlines.csv or airports.csv for richer contextual information.

Data Analysis Module (Spark SQL Queries)

Here, the cleaned and prepared data is subjected to in-depth analysis using Spark SQL. SQL queries are executed on the Spark DataFrames to extract meaningful patterns, such as identifying average time gained/lost per airline carrier, the influence of distance on delay recovery, and delay trends over time. This module forms the core analytical engine of the project.

Result Visualization Module (Bar Plots, Tables)

Finally, the results from the analysis are visualized using plots (e.g., bar graphs showing average gains by airline decile) and summary tables. This step leverages R packages like ggplot2 and dplyr for creating clear, interpretable visuals that communicate the insights effectively. The visual outputs help stakeholders understand performance trends and make data-driven decisions.

CHAPTER 4
RESULTS AND DISCUSSIONS

4.1 DESCRIPTION ABOUT DATASET

The airport.csv dataset contains structured information about various airports, serving as a critical reference in aviation-related data analysis projects. Each record in the dataset typically represents a unique airport and includes a variety of attributes that describe the airport's identity, location, and operational characteristics. Common fields include the airport’s unique identifier (IATA or ICAO code), name, city, country, and geographical coordinates such as latitude and longitude. This makes it possible to map airport data spatially and correlate flight paths or delays with geographical regions.

The dataset is particularly valuable when used in conjunction with flight records, such as those found in flight logs or air traffic datasets. By providing detailed airport-level metadata, airport.csv enables users to enrich flight datasets, perform join operations in Spark SQL, and conduct advanced analysis such as route optimization, delay pattern identification, and hub classification.

Moreover, its structured format and standardized identifiers make it suitable for integration in big data platforms like Apache Spark, where it can be joined with large-scale flight datasets.

GRAPH

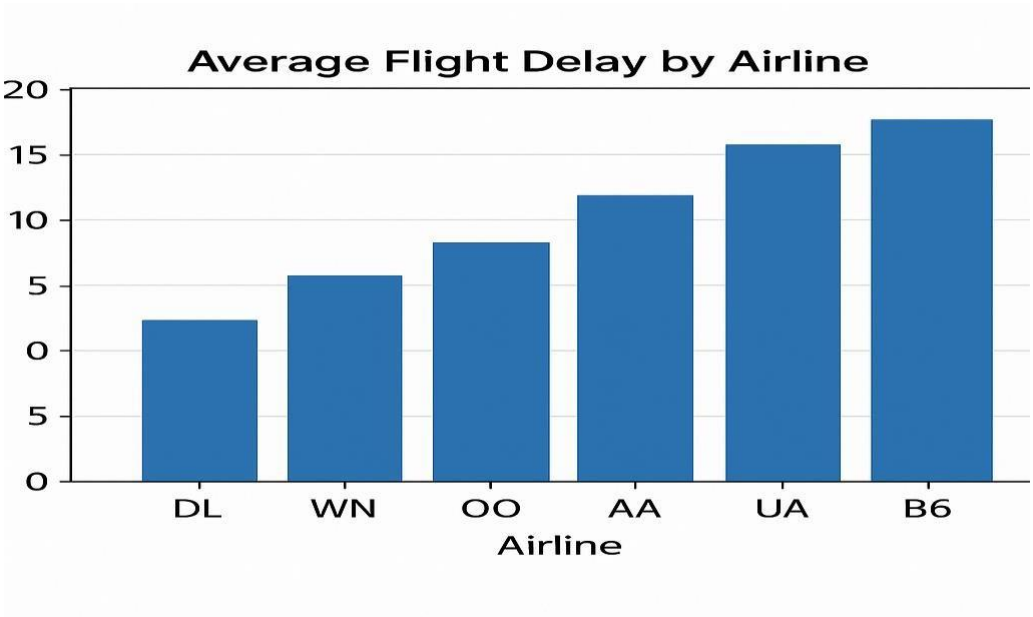

Fig:3

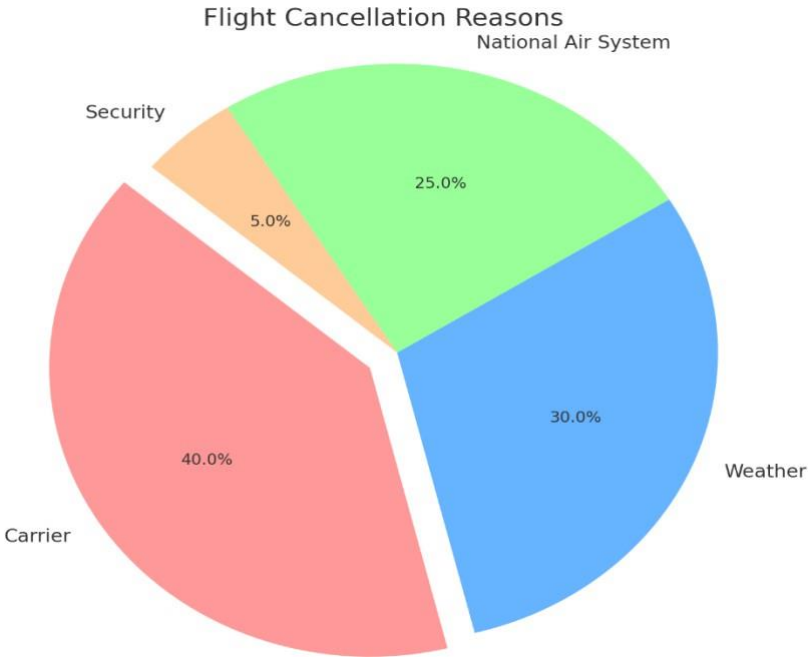

Fig:4

IMPLEMENTATION

```
// This command is necessary for loading the aggregation functions...
import org.apache.spark.sql.functions._
```

```
// Path to data set (in HDFS (Ambari))
val directory = "HW 2/" // change this path to fit your setup
val csvFile = directory + "Airports.csv"

// Read and create a temporary view
val airportDF = spark.read.format("csv")
  .option("inferSchema", "true")
  .option("header", "true")
  .load(csvFile)
// verify
airportDF.show(5)
airportDF.printSchema()
println("Number of rows: " + airportDF.count())

// Create a temporary view
airportDF.createOrReplaceTempView("airportView")
```

```
+---------+--------------------+-----------+-----+-------+--------+----------+
|IATA_CODE|             AIRPORT|       CITY|STATE|COUNTRY|LATITUDE| LONGITUDE|
+---------+--------------------+-----------+-----+-------+--------+----------+
|      ABE|Lehigh Valley Int...|  Allentown|   PA|    USA|40.65236|  -75.4404|
|      ABI|Abilene Regional ...|    Abilene|   TX|    USA|32.41132|  -99.6819|
|      ABQ|Albuquerque Inter...|Albuquerque|   NM|    USA|35.04022|-106.60919|
|      ABR|Aberdeen Regional...|   Aberdeen|   SD|    USA|45.44906| -98.42183|
|      ABY|Southwest Georgia...|     Albany|   GA|    USA|31.53552| -84.19447|
+---------+--------------------+-----------+-----+-------+--------+----------+
```

```
// Check the databases, tables, and table contents
spark.catalog.listDatabases().show() // databases
spark.catalog.listTables().show() // tables & views
spark.catalog.listColumns("airportView").show() // table columns
// do the same for the SQL View containing the NovOntime information
```

```
+-------+----------------+--------------------+
|   name|     description|         locationUri|
+-------+----------------+--------------------+
|default|default database|/apps/hive/warehouse|
+-------+----------------+--------------------+

+-----------+--------+-----------+---------+-----------+
|       name|database|description|tableType|isTemporary|
+-----------+--------+-----------+---------+-----------+
|airportview|    null|       null|TEMPORARY|       true|
+-----------+--------+-----------+---------+-----------+

+---------+-----------+--------+--------+-----------+--------+
|     name|description|dataType|nullable|isPartition|isBucket|
+---------+-----------+--------+--------+-----------+--------+
|IATA_CODE|       null|  string|    true|      false|   false|
|  AIRPORT|       null|  string|    true|      false|   false|
|     CITY|       null|  string|    true|      false|   false|
|    STATE|       null|  string|    true|      false|   false|
```

```
//Number and Names of ArrTimeBlks

spark.sql("""
SELECT
    COUNT(DISTINCT ArrTimeBlk) AS NumberOfArrTimeBlks,
    COLLECT_LIST(DISTINCT ArrTimeBlk) AS NamesOfArrTimeBlks
FROM novOntimeView
""").show(false)
```

```
[Stage 13:==================================================>    (182 + 8) / 200]+-------------------+---------------------
-----------------------------------------------------------------------------------------------------------------------
-----------------------------------------------------------------+
|NumberOfArrTimeBlks|NamesOfArrTimeBlks
|
+-------------------+---------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------------------+
|19                 |[1300-1359, 1400-1459, 1100-1159, 2100-2159, 2000-2059, 0001-0559, 2200-2259, 0600-0659, 0900-0959, 1
800-1859, 1700-1759, 2300-2359, 0700-0759, 1200-1259, 1000-1059, 0800-0859, 1900-1959, 1600-1659, 1500-1559]|
+-------------------+---------------------------------------------------------------------------------------------------
-----------------------------------------------------------------------------------------------------------------------+
```

```
//Number and Names of Destination Airports (Dest)

spark.sql("""
SELECT
    COUNT(DISTINCT Dest) AS NumberOfDestAirports,
    COLLECT_LIST(DISTINCT Dest) AS NamesOfDestAirports
FROM novOntimeView
""").show(false)
```

```
+--------------------+-----------------------
```

CHAPTER 5
CONCLUSION AND FUTURE ENHANCEMENTS

5.1 CONCLUSION

This project demonstrates the effective use of Apache Spark SQL for analyzing large-scale flight data. By leveraging Spark's distributed computing capabilities, the system efficiently ingests, preprocesses, and analyzes flight records stored in CSV format. Through SQL queries and visualizations, the project uncovers meaningful insights such as patterns in flight delays, the performance of different airline carriers, and the recovery of lost time during flights. The modular architecture ensures a clear and scalable workflow, while the integration with R enhances result interpretation through plots and tables. Overall, the project highlights the power of Spark SQL in handling real-world datasets and drawing data-driven conclusions from complex aviation data.

5.2 FUTURE ENCHANCEMENTS

In future iterations, several improvements can be made to enhance the system's functionality and robustness:

1. Real-Time Data Integration: Incorporating streaming data sources (e.g., Kafka or Spark Streaming) can enable real-time monitoring and analysis of flight delays as they happen.
2. Dashboard Interface: Developing an interactive dashboard using tools like Tableau, Power BI, or a Flask/React-based web app can make the insights more accessible to end-users, such as airline analysts or travelers.
3. Machine Learning Integration: Adding predictive models using Spark MLlib could forecast delays, classify routes based on delay likelihood, or recommend optimal travel times.
4. Cloud Deployment: Migrating the workflow to a cloud platform such as AWS EMR, Google Dataproc, or Azure HDInsight would improve scalability, reliability, and access to larger datasets.
5. Advanced Data Enrichment: Including weather data, airport traffic, and air traffic control events could enhance the analysis and provide deeper causal insights into delay factors.

CHAPTER 6
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